

Title Page

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Protocol Number: ImPRESS-losartan study

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SIGNATURE PAGE

I hereby acknowledge this SAP and that its contents describes the primary analyses to be performed in the ImPRESS trial. I have had no knowledge of any unblinded results by the time of signature. I also acknowledge that any substantial changes in analyses after I have signed the final first version of this SAP will be denoted post-hoc analyses.

SPONSOR REPRESENTATIVE:

Kyrre Eeg Emblem, PhD, Head of department
Department of Physics and Computational
Radiology, Division of Radiology and Nuclear
Medicine, Oslo University Hospital

Kyrre Eeg Emblem

2026-06-24

Signature

Date (dd/mmm/yyyy)

**PRINCIPAL/COORDINATING
INVESTIGATOR:**

Petter Brandal, senior consultant and professor
Department of Oncology, Oslo University
Hospital; Institute of Clinical Medicine,
University of Oslo

Petter Brandal

2026-06-24

Signature

Date (dd/mmm/yyyy)

TRIAL STATISTICIAN:

Inge Christoffer Olsen, PhD
Department of Research Support for Clinical
Trials, Oslo University Hospital, Oslo, Norway.

Inge Christoffer Olsen

2026-06-24

Signature

Date (dd/mmm/yyyy)

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Version History

This statistical analysis plan (SAP) for the ImPRESS study is based on the protocol version 8.0 dated 14-NOV-2025.

SAP Version	Date	Change	Rationale
1.0	24-JUN-2026	Not applicable	Original version

List of Abbreviations

Abbreviation	Definition
ADC	apparent diffusion coefficient
AIC	Akaike Information Criterion
AN	newly diagnosed glioblastoma cohort (study cohort code)
AR	recurrent glioblastoma cohort (study cohort code)
BM	brain metastases cohort (study cohort code)
CPT	Common Protocol Template
CSR	Clinical Study Report
CTU	Clinical Trials Unit
DSC	dynamic susceptibility contrast (MRI perfusion)
ECOG	Eastern Cooperative Oncology Group
ED50	dose producing half of the maximal effect (Emax models)
EORTC	European Organisation for Research and Treatment of Cancer
FAS	Full Analysis Set
FLAIR	fluid-attenuated inversion recovery (MRI)
ICH	International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use
ICH E9(R1)	ICH addendum on estimands and sensitivity analysis in clinical trials
ICMJE	International Committee of Medical Journal Editors
IMP	investigational medicinal product
KPS	Karnofsky Performance Scale
MAR	missing at random
MCP-Mod	Multiple Comparison Procedure and Modelling
MedDRA	Medical Dictionary for Regulatory Activities
MR	magnetic resonance
MRI	magnetic resonance imaging
NANO	Neurologic Assessment in Neuro-Oncology
OSF	Open Science Framework
QLQ-BN20	EORTC Quality of Life Questionnaire, Brain Neoplasm 20-item module
QLQ-C30	EORTC Quality of Life Questionnaire, Core 30-item
RANO	Response Assessment in Neuro-Oncology
ROI	region of interest (ROI2 = oedema-targeted ROI)
SAF	Safety Analysis Set
SAP	Statistical Analysis Plan

Biomarker panel analytes

Abbreviation	Definition
ANG-1, ANG-2	angiopoietin-1, angiopoietin-2
ANGPTL4	angiopoietin-like 4
CA9	carbonic anhydrase 9
CCL2	C-C motif chemokine ligand 2
CD31	cluster of differentiation 31 (PECAM-1)
CD163	cluster of differentiation 163
CX3CL1	C-X3-C motif chemokine ligand 1 (fractalkine)
CXCL9–11	C-X-C motif chemokine ligands 9 to 11
EGF	epidermal growth factor
ErbB2	erb-b2 receptor tyrosine kinase 2 (HER2)
FGF	fibroblast growth factor
GITR/TNFRSF18	glucocorticoid-induced TNFR-related protein / TNF receptor superfamily member 18
IFN-gamma	interferon gamma
IGFBP7	insulin-like growth factor-binding protein 7
IL-1beta, IL-2, IL-6, IL-8, IL-12	interleukin-1 beta, -2, -6, -8, -12
MMP-10	matrix metalloproteinase-10
PDGF	platelet-derived growth factor
PlGF	placental growth factor
TGF-alpha	transforming growth factor alpha
THBS2	thrombospondin-2
TNF-alpha	tumour necrosis factor alpha
TNFSF6	TNF superfamily member 6 (Fas ligand)
VEGF-A, VEGF-C	vascular endothelial growth factor A, C
VEGFR2	vascular endothelial growth factor receptor 2

1. Introduction

The purpose of this Statistical Analysis Plan is to detail the statistical principles, analyses methods, and data handling conventions for the ImPRESS trial. The ImPRESS trial is a phase 2 randomised, stepped-wedge, assessor-blinded dose-finding trial evaluating losartan as an adjunct to standard of care. It is based the structure and conventions of the TranCelerate SAP Template v5 and corresponds to the analysis framework as defined in the ImPRESS protocol version 8.0 dated 14 November 2025.

The protocol is described for three different patient populations: recurrent glioblastoma (AR), newly diagnosed glioblastoma (AN) and brain metastases (BM). This SAP will only consider the newly diagnosed glioblastoma population; separate plans will cover the other populations. Changes to the protocol-planned analyses are described in Section 10.

1.1. Trial Objectives and Associated Estimands

1.2. Primary Objective and Associated Estimands

1.2.1. Primary Objective

The primary objective is to assess the dose-response relationship of the angiotensin II receptor inhibitor losartan on brain tumour perfusion and mechanical solid stress in patients with newly diagnosed glioblastoma.

1.2.2. Primary estimand

The primary estimand is described by the following attributes in the table below:

Estimand Characteristic	Description
Population	Patients newly diagnosed with glioblastoma
Treatment	Intervention for 14 days with different doses of losartan: 0 mg, 25 mg, 50 mg, 100 mg, regardless of other treatment or if the treatment was given fully or partially according to protocol intention.
Variable of interest	1) Change from baseline to 14 days after treatment start in perfusion as measured by the radiographic biomarker relative cerebral blood flow in the primary tumour area 2) Change from baseline to 14 days after treatment start in mechanical solid stress as measured by the radiographic biomarker tissue stiffness as derived from the shear wave modulus G in the MR-elastography readout in the primary tumour area Both endpoints are normalised to reference tissue.
Population-level summary	Mean change from baseline for each dose

Intercurrent Events	Strategy
Discontinuation of investigational trial intervention for any reason	Treatment policy strategy (addressed in treatment conditions attribute)
Initiation of rescue medication or change in background medication (dose and product)	Treatment policy strategy (addressed in treatment conditions attribute)
Death	Hypothetical strategy, i.e. what would the outcome have been if death had not occurred. Outcomes after this event is handled as missing data.

1.2.3. Sensitivity estimands

A sensitivity estimand for the primary objective is defined identical to the primary estimand but where the endpoint is measured in the oedema-targeted area.

1.3. Secondary Objectives and Associated Estimands

1.3.1. Secondary Objective 1

The first secondary objective is to assess the dose-response relationship of the angiotensin II receptor inhibitor losartan on conventional and experimental radiographic characteristics in patients with newly diagnosed glioblastoma.

We divide this objective into a short- and long-term treatment length assessment of the dose-response relationship. The short-term treatment effect is the causal effect of 14 days treatment with the different doses on the outcome. The long-term treatment effect is the causal effect of more than 42 days of treatment with the different doses on the outcome.

1.3.1.1. Secondary Objective 1 short-term treatment estimands

The estimands are identical to the primary estimand, but with different endpoints. The endpoints for these estimands are change from baseline to 14 days after treatment initiation in

- DSC K2 leakage
- relative DSC extravasation
- relative apparent diffusion coefficient
- relative cerebral blood
- relative direction value
- relative fractional anisotropy
- relative restricted spectrum imaging
- relative transit time
- relative vessel caliber
- relative vessel size index
- relative vortex area
- tissue elasticity
- tissue relative viscosity

- tissue stiffness
- tissue viscosity
- total volume of contrast agent enhancing lesion on T1-weighted MRI in cm³
- total volume of pathologic signal on T2/FLAIR-weighted MRI in cm³

1.3.1.2. Secondary Objective 1 long-term treatment estimands

The estimands are identical to the primary and secondary estimands with the following specific alterations and additions:

Treatment	Intervention for more than 42 days with different doses of losartan: 0 mg, 25 mg, 50 mg, 100 mg, regardless of other treatment or if the treatment was given fully or partially according to protocol intention.
Variable of interest	Change from baseline to 140 days after treatment start in all radiographic endpoints as defined for the primary and secondary short treatment estimands
Additional intercurrent event	Strategy
Short-term (42 days or less) treatment with losartan of any dose	Treatment policy strategy

1.3.2. Secondary Objective 2

The second secondary objective is to assess the dose-response relationship of losartan as add-on to standard treatment on clinical or patient reported endpoints in patients with newly diagnosed glioblastoma.

We divide also this objective into a short- and long-term treatment length assessment of the dose-response relationship. The short-term treatment effect is the causal effect of 14 days treatment with the different doses on the outcome. The long-term treatment effect is the causal effect of more than 42 days of treatment with the different doses on the outcome.

1.3.2.1. Secondary Objective 2 short-term treatment estimands

The estimands are identical to the short-term radiographic estimands but with different endpoints. The endpoints for these estimands are change from baseline to 14 days after treatment initiation in

- Karnofsky Performance Scale
- Eastern Cooperative Oncology Group (ECOG) score
- Neurologic Assessment in Neuro-Oncology (NANO) scores

In addition, there is an estimand on steroid use with a dichotomous endpoint defined as “Any use of steroids”. The population-level summary for this estimand is the proportion.

1.3.2.2. Secondary Objective 2 long-term treatment estimands

The estimands are identical to the long-term treatment radiographic estimands but with different endpoints. The endpoints for these estimands are change from baseline to 154 and 238 days after treatment initiation in

- Karnofsky Performance Scale
- Eastern Cooperative Oncology Group (ECOG) score
- Neurologic Assessment in Neuro-Oncology (NANO) scores
- European Organisation for Research and Treatment of Cancer (EORTC) Quality of Life Questionnaire QLQ-C30
- EORTC QLQ-BN20

In addition, there are ordinal endpoints according to the Response Assessment in Neuro-Oncology (RANO) criteria 140 and 224 days after treatment initiation: one for each timepoint and one best overall response. The population-level summary is the proportional odds ratio for each dose level.

Finally, there are four estimands concerning progression-free and overall survival of long-term treatment. The estimands are identical to the long-term treatment radiographic estimands but with different endpoints and population-level summary. The endpoints are defined as

- Time to progression in the period from treatment start until six months
- Time to progression in the period from treatment start until 24 months
- Time to death in the period from treatment start until 12 months
- Time to death in the period from treatment start until 24 months

The population level summary for all these endpoints is the hazard ratio of each dose against no treatment (0 mg).

1.3.3. Secondary Objective 3

The third secondary objective is to assess the dose–response relationship of the angiotensin II receptor inhibitor losartan on circulating blood and serum biomarkers of angiogenesis, vascular function, and inflammation in patients with newly diagnosed glioblastoma.

As for the radiographic and clinical objectives, this objective is divided into a short- and long-term treatment-length assessment of the dose–response relationship. The short-term treatment effect is the causal effect of 14 days treatment with the different doses on the outcome. The long-term treatment effect is the causal effect of more than 42 days of treatment with the different doses on the outcome.

The biomarker panel comprises the analytes specified in the protocol, measured from blood/serum samples using multiplex immunoassay (Bio-Plex / Luminex) panels:

- Vascular endothelial growth factors and receptors:
 - VEGF-A, VEGF-C, VEGFR2, CD31, FGF, PDGF, IGFBP7
- Angiopoietins:
 - ANG-1, ANG-2, ANGPTL4
- Placental growth factor:
 - PlGF
- Epidermal growth factor receptors:
 - EGF, TGF- α
- Extracellular matrix and stromal markers:
 - hyaluronan, collagen IV α 1, MMP-10, Tenascin C, CD163, CA9, THBS2
- Inflammatory cytokines:
 - CXCL9–11, CCL2, CX3CL1, ErbB2, Fas Ligand/TNFSF6, GITR/TNFRSF18, TNF- α , Granzyme B, IFN- γ , IL-1 β , IL-2, IL-6, IL-8, IL-12

All biomarker endpoints are continuous concentrations (pg/mL). Given the number of analytes and the exploratory phase 2 setting, these analyses are interpreted descriptively and no multiplicity adjustment is applied. Prior to unblinding model fit was assessed by comparing Akaike's Information Criterion of the primary model fitted with raw and log scale. For all analytes the log scale gave the best fit and is therefore reported. A separate report (ImPRESS – Biomarker Model Fit Assessment, appendix) reports the model fit assessments.

1.3.3.1. Secondary Objective 3 short-term treatment estimands

The estimands are identical to the short-term radiographic estimands (Secondary Objective 1) but with different endpoints. The endpoints are the change from baseline to the post-baseline assessments within the short-term period (Days 15 and 43) in the concentration of each biomarker analyte listed above transposed to the natural logarithmic scale. Baseline is the screening sample obtained prior to initiation of losartan treatment. The population-level summary is the mean log change from baseline for each dose.

1.3.3.2. Secondary Objective 3 long-term treatment estimands

The estimands are identical to the long-term treatment radiographic estimands (Secondary Objective 1) but with different endpoints. The endpoints are the change from baseline to the long-term assessment (Day 225) in the concentration of each biomarker analyte transposed to the natural logarithmic scale. The population-level summary is the mean log change from baseline for each long-term treatment group.

1.3.4. Secondary Safety Objective

This objective is to assess the safety of losartan treatment in patients with newly diagnosed glioblastoma.

1.3.4.1. Safety Objective Estimands

The safety objective estimands is described by the following attributes in the table below:

Estimand Characteristic	Description
Population	Patients newly diagnosed with glioblastoma
Treatment	Intervention with different doses of losartan: 0 mg, 25 mg, 50 mg, 100 mg, regardless of other treatment or if the treatment was given fully or partially according to protocol intention.
Endpoint	1) Occurrence of adverse events 2) Systolic and diastolic blood pressure 3) Pulse 4) Conventional laboratory blood parameters
Population-level summary	1) Descriptive: Number and proportion of participants with adverse events by MedDRA preferred term and system organ class 2) Mean change from baseline for each dose
Intercurrent Events	Strategy
Discontinuation of investigational trial intervention for any reason	Treatment policy strategy (addressed in treatment conditions attribute)
Initiation of rescue medication or change in background medication (dose and product)	Treatment policy strategy (addressed in treatment conditions attribute)
Death	While-alive strategy

1.4. Exploratory Objective

An exploratory objective is to further explore and magnetic resonance imaging methods and their correlation to clinical and blood/serum data with the objective to identify personalized treatment options in patients with newly diagnosed glioblastoma.

There will be no pre-specified estimands for this objective. The objective will be explored using a wide variety of methods and approaches.

1.5. Study Design

This part of the ImPRESS study is a phase 2 exploratory, open-label, blinded assessor, single center, multi-dose individual-randomized stepped-wedge trial on patients with newly diagnosed glioblastoma.

The randomised treatment allocation will follow a prespecified randomised scheme according to the stepped wedge design. To assess any long-term beneficial effects of the IMP, some patients will receive the IMP for a longer period. Participants in groups AN3_1, AN3_2 and AN 3_3 will all receive extended IMP treatment, whereas patients in groups AN1_1, AN1_2, AN1_3, AN2_4, AN2_5 and AN2_6 will be randomized to short or extended treatment with the IMP. This leads to a distribution with 1/3 of patients with no extended IMP treatment, 1/3 extended lower doses of IMP (1/6 25 mg and 1/6 50 mg), and 1/3 extended high dose of IMP. The treatment scheme is presented in .

Group	IMP dose	Baseline	Days 1 to 14	Days 15 to 28	Days 29 to 44	Days 45 to 238	Days > 238
AN1_1	25 mg						
AN1_2	25 mg						
AN1_3	25 mg						
AN2_4	50 mg						
AN2_5	50 mg						
AN2_6	50 mg						
AN3_1	100 mg						
AN3_2	100 mg						
AN3_3	100 mg						

Figure 1 Randomisation scheme

2. General Considerations

2.1. Statistical Hypotheses

The ImPRESS study is a phase 2 trial designed to explore the dose-response relationship of losartan on primarily imaging-based biomarkers in patients with brain tumours. Thus, it is not a confirmatory study, and all hypothesis testing should be considered supportive and not confirmatory.

The null hypothesis for each of the two primary endpoints is that there is no dose-response relationship across the planned dose levels (0, 25, 50, 100 mg).

Unless otherwise stated, statistical tests are two-sided with a nominal 5% significance level and 95% confidence intervals.

2.2. Multiplicity Adjustment

While it is an exploratory study not designed for confirmatory claims, we will make some efforts to control the family-wise type 1 error:

- The estimands are divided into primary and secondary estimands with more type 1 error control in the primary estimands
- For the assessment of dose-response for the primary estimands we will adjust for multiplicity according to the Multiple Comparison Procedure and Modelling (MCP-Mod) framework as detailed in section 4.1.
- No further multiplicity adjustments will be done.

2.3. Impact of Intercurrent Events Strategies

Intercurrent events are handled in accordance with the estimand definitions specified in Section 1. For most estimands, a **treatment policy strategy** is used. Under this strategy, participants are analysed according to the treatment they were randomised to receive, regardless of treatment adherence, treatment interruptions, dose modifications, or changes in concomitant or background therapies occurring after randomisation.

For estimands evaluating short-term treatment effects, participants who discontinue the investigational medicinal product prior to completion of the planned short-term treatment period remain part of the estimand target population, and their outcomes are interpreted as reflecting the effect of the assigned treatment strategy rather than the effect of treatment actually received.

For long-term treatment estimands, short-term treatment with losartan (defined as treatment duration of 42 days or less, irrespective of dose) is treated as an intercurrent event and handled using a treatment policy strategy. Under this approach, outcomes observed in participants who discontinue losartan prior to the long-term assessment window are interpreted as part of the assigned long-term treatment strategy, even if the intended long-term exposure was not achieved.

This strategy implicitly assumes that any short-term exposure to losartan does not confer a sustained long-term benefit on the outcomes of interest in the absence of continued treatment. As a result, the estimated long-term treatment effects reflect the effect of assignment to a long-term losartan treatment strategy rather than the effect among participants who successfully maintained

long-term treatment. This approach is considered conservative, as participants assigned to the no long-term losartan strategy (0 mg) may have received short-term losartan treatment, potentially improving outcomes in this group and thereby attenuating estimated long-term treatment effects.

For estimands where death occurs prior to the assessment timepoint, different strategies are applied depending on the objective. For primary and secondary efficacy estimands evaluating radiographic and clinical outcomes, a **hypothetical strategy** is used, targeting the outcome that would have been observed had death not occurred prior to the assessment timepoint. For safety estimands, a **while-alive strategy** is applied, such that outcomes are defined and summarised only for the period during which participants are alive.

2.4. Handling of Missing Data

Missing data may arise due to missed visits, early discontinuation of study procedures, withdrawal of consent, or death. The handling of missing data is aligned with the estimand strategies defined in Section 1 and the intercurrent event handling described in Section 2.3.

For primary and secondary efficacy estimands based on continuous radiographic, clinical and patient reported endpoints, missing outcome data will be handled implicitly through likelihood-based mixed effects models fitted using maximum likelihood estimation. Under this approach, estimation is valid under the assumption that data are missing at random (MAR), conditional on observed data included in the model.

No explicit imputation of missing values will be performed for these analyses. All available post-baseline data contributing to the relevant estimand will be used, consistent with the treatment policy or hypothetical strategies specified for each estimand.

The MAR assumption is recognised as a strong and untestable assumption. However, given the exploratory phase 2 nature of the ImPRESS trial and the primary focus on estimation and characterisation of dose–response relationships rather than confirmatory hypothesis testing, this approach is considered appropriate. Results will therefore be interpreted cautiously, with emphasis on effect estimates and uncertainty rather than formal statistical significance.

For time-to-event endpoints, incomplete follow-up will be handled through right censoring at the last time the participant is known to be event-free, under the standard assumption of non-informative censoring.

No formal sensitivity analyses specifically targeting departures from the MAR assumption are planned.

3. Analysis Sets

Two analysis sets will be defined for the ImPRESS trial.

3.1. Full Analysis Set (FAS)

The Full Analysis Set includes all participants with newly diagnosed glioblastoma who were randomised to a treatment scheme.

Participants will be analysed according to the treatment strategy to which they were randomised, irrespective of treatment adherence, treatment interruptions, dose modifications, or early discontinuation of the investigational medicinal product, in accordance with the treatment policy strategy defined for the primary and secondary estimands.

The Full Analysis Set will be used for all efficacy analyses.

3.2. Safety Analysis Set (SAF)

The Safety Analysis Set includes all participants who received at least one dose of the investigational medicinal product.

Participants in the Safety Analysis Set will be analysed according to the treatment actually received. The Safety Analysis Set will be used for the safety analyses.

4. Analyses Supporting Primary Objective

4.1. Primary Estimands

4.1.1. Definition of Endpoints

The ImPRESS trial has two co-primary estimands corresponding to two imaging-based efficacy outcomes in participants with newly diagnosed glioblastoma.

The first primary estimand targets the **mean change from baseline to 14 days after treatment initiation in tumour perfusion**, defined as relative cerebral blood flow measured in the primary tumour area using magnetic resonance imaging–based perfusion techniques and normalised to reference tissue.

The second primary estimand targets the **mean change from baseline to 14 days after treatment initiation in mechanical solid stress**, defined as tumour tissue stiffness derived from the shear wave modulus (G) obtained from MR elastography in the primary tumour area and normalised to reference tissue.

For both primary estimands, baseline is defined as the assessment obtained prior to initiation of losartan treatment. Change from baseline is calculated as the post-baseline value minus the baseline value.

No transformation of the underlying variables (e.g. logarithmic or square-root transformation) is planned prior to analysis; the estimands are defined on the original measurement scales of the respective biomarkers.

Details of image acquisition, processing, and quality control procedures are described in the study protocol.

4.1.2. Main Analytical Approach

The main analysis targets the primary estimands defined in Section 1.2 and follows the ICH E9(R1) estimand framework. Analyses will be performed on the Full Analysis Set (Section 3).

For each co-primary estimand, the model will use all outcomes from the post-baseline visits within the short-term treatment period (i.e. Day 15, 29 and 43). The primary analytical model is a repeated-measures linear mixed effects model with a subject-specific random intercept to account for within-participant correlation across visits. Fixed effects include visit (categorical) and randomised treatment dose at the visit. Further specifications as below:

- For descriptive longitudinal summaries across early visits, estimated marginal means by visit and treatment start step will be obtained from a mixed model of the observed outcome including baseline as a covariate and a step-by-visit interaction, with subject-specific random intercepts.

- For the dose-response analysis, mean change from baseline will be estimated for each planned dose group using a mixed model including baseline as a covariate, visit, and dose group, with a subject-specific random intercept. Model-based estimates and their covariance matrix will be used as input to the MCP-Mod procedure.

Specifically, the mixed model is defined as:

$$\Delta P_{ij} = A_i + \mu_{trt} + \mu_j + \epsilon_{ij}$$

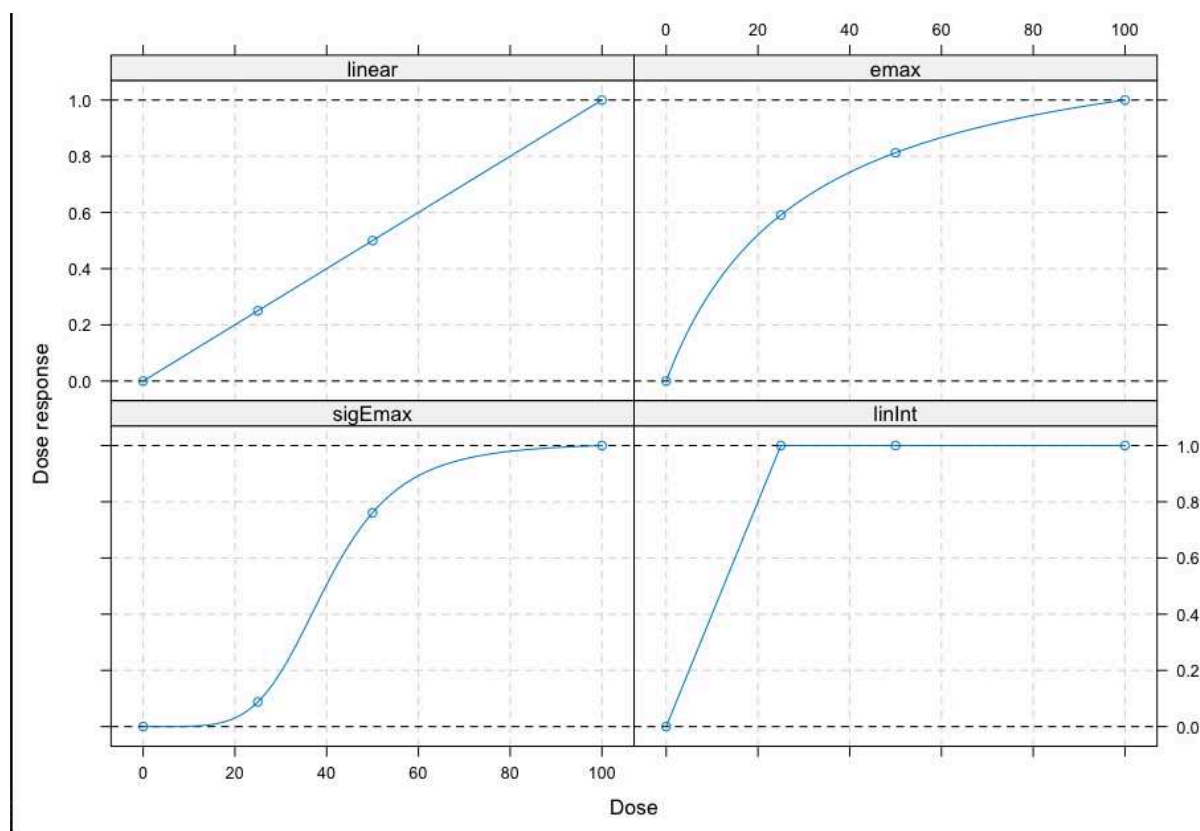
where ΔP_{ij} is the change from baseline of patient $i = 1..N$ at timepoint j , A_i is the random intercept of patient i , μ_{trt} is the treatment at timepoint j , μ_j is the timepoint j , and ϵ_{ij} is the residual

No adjustments for additional baseline characteristics are planned beyond the baseline value of the endpoint.

4.1.2.1. Dose–response assessment (MCP step)

The dose–response relationship will be evaluated using the Multiple Comparison Procedure and Modelling (MCP-Mod) framework with a pre-specified set of candidate dose–response models. The candidate models are:

1. A linear model
2. A **piecewise linear (linear spline) model** with knots fixed at the planned dose levels
3. An Emax model with ED50 fixed at 30 mg
4. A sigmoid Emax model with ED50 fixed at 40 mg and steepness fixed at 5



For each candidate model, optimal contrast tests will be constructed using the model-based dose-group estimates and their covariance matrix. Multiplicity adjustment will be applied across the set of candidate models.

A one-sided hypothesis test with a nominal significance level of 2.5% will be used to test the null hypothesis of no dose-response relationship against the alternative that the dose-response follows one of the pre-specified candidate models. A statistically significant dose-response signal is declared if at least one multiplicity-adjusted p-value is below the nominal significance level.

4.1.2.2. Modelling step (Mod step)

If a statistically significant dose-response signal is established in the MCP step, the corresponding dose-response models will be fitted to estimate the dose-response relationship. Model-based estimates, confidence intervals, and relevant dose metrics (e.g. effective doses) will be derived and presented descriptively.

4.1.2.3. Multiplicity across co-primary endpoints

Multiplicity across the two co-primary endpoints will be addressed using a hierarchical testing strategy. The MCP-Mod procedure will first be applied to the perfusion endpoint. Only if a statistically significant dose-response signal is established for perfusion will the same procedure be applied to the solid stress endpoint. If the perfusion endpoint does not demonstrate a statistically significant dose-response signal, the solid stress endpoint will be regarded as not statistically significant within this hierarchical framework. Note that the hierarchical rule will be applied at interpretation of the results, the statistical inference will be performed regardless of interpretation, i.e. all hypothesis tests will be performed.

4.1.2.4. Intercurrent events and missing data

Intercurrent events are handled according to the strategies specified in Sections 1 and 2.3. Missing outcome data are handled implicitly through likelihood-based estimation in the mixed effects models under the assumption that data are missing at random, as described in Section 2.4. No explicit imputation is performed for the primary analyses.

4.1.2.5. Presentation of results

Results will be presented as model-based estimates of mean change from baseline for each dose group with corresponding 95% confidence intervals. For the MCP step, multiplicity-adjusted p-values will be reported for each candidate dose–response model. Graphical displays of the candidate models and fitted dose–response curves will be provided to aid interpretation.

4.1.2.6. Model diagnostics

Model assumptions will be assessed using standard diagnostic procedures, including visual inspection of residual distributions and residuals versus fitted values. These diagnostics are intended to evaluate the adequacy of the assumed model structure. No data-driven modifications of the primary analysis model are planned based on these assessments.

4.1.3. Sensitivity Analyses

For each co-primary estimand, a sensitivity analysis will be performed using an alternative definition of the imaging endpoint based on an oedema-targeted region of interest (ROI2). The sensitivity analyses differ from the primary analyses only in the region of interest used to derive the endpoint; all other estimand attributes and analytical methods remain unchanged.

Results from the sensitivity analyses will be interpreted in relation to the primary analyses to assess robustness to the choice of region of interest.

4.1.4. Supplementary Analyses

Supplementary analyses will include descriptive and supportive summaries to aid interpretation of the primary analyses. These include:

- Model-based estimated mean change from baseline by dose group with 95% confidence intervals. This is based on the primary analysis model.
- Pairwise differences in mean change from baseline comparing each active dose to 0 mg as mean with 95% confidence intervals. This is based on the primary analysis model.
- Graphical displays of observed and model-based responses over time by visit and treatment start step.
- Graphical presentation of candidate dose–response models and fitted dose–response curves from the Mod step.

Finally a contrast comparing **any dose of losartan** (25 mg, 50 mg, or 100 mg) versus **no losartan (0 mg)** will be performed. For this analysis, all active dose groups will be pooled and compared with the 0 mg group using the same linear mixed effects model as used for the primary analyses, including baseline adjustment and subject-specific random intercepts.

The estimated mean difference in change from baseline between any losartan dose and no losartan will be reported with a 95% confidence interval. This analysis is intended to provide a descriptive assessment of a potential common treatment effect of losartan, corresponding to a step-like dose–response pattern, and is not part of the multiplicity-controlled MCP-Mod procedure.

5. Analyses Supporting Secondary Objectives

5.1. Secondary Estimands

Secondary estimands address dose–response relationships of losartan on additional imaging, clinical, and patient-reported outcomes in participants with newly diagnosed glioblastoma, assessed over short-term and long-term periods following treatment initiation, as defined in Section 1.3. All secondary analyses are exploratory and will be performed on the Full Analysis Set (Section 3).

For short-term estimands, outcomes are operationalised using scheduled post-baseline assessments within the short-term period (nominally Days 15, 29, and 43) and the estimand-relevant effect is derived from the first post-baseline assessment. For long-term estimands, outcomes are operationalised using scheduled long-term assessments (nominally Days 155 and 239, and where relevant Day 141 and/or Day 225).

No multiplicity adjustment is planned for secondary endpoints.

5.2. Secondary Imaging Endpoints

Secondary imaging endpoints will be analysed using linear mixed effects models analogous to those used for the primary imaging endpoints. Models will include baseline adjustment and subject-specific random intercepts and will use scheduled post-baseline imaging assessments within the relevant short-term or long-term period. Model-based estimates by dose group will be presented with 95% confidence intervals.

5.3. Secondary Clinical and Patient-Reported Endpoints

5.3.1. Neurological endpoints (KPS, ECOG, NANO)

5.3.1.1. Short-term analyses (randomised dose during short-term period)

For neurological endpoints collected during the short-term period (KPS, ECOG, NANO), analyses will use all available post-baseline measurements at nominal Days 15 and 43. A linear mixed effects model with subject-specific random intercept will be fitted, including baseline value of the endpoint, nominal visit (categorical), and randomised dose group as fixed effects. Model-based estimates of mean change from baseline by dose group will be reported with 95% confidence intervals. Pairwise differences versus 0 mg (25/50/100 mg vs 0 mg) will be presented as supportive summaries with 95% confidence intervals.

5.3.1.2. Long-term analyses (randomised long-term treatment strategy).

For long-term neurological endpoints (nominal Days 155 and 239), analyses will be based on the long-term treatment assignment and will use a linear mixed effects model with subject-specific random intercept. The model will include baseline value of the endpoint and a treatment-by-visit interaction to allow treatment effects to differ by visit. Model-based estimates by long-term treatment group and visit will be reported with 95% confidence intervals, together with pairwise differences versus the 0 mg long-term treatment group.

5.3.2. Quality of life endpoints (EORTC QLQ-C30, QLQ-BN20)

Quality of life endpoints (QLQ-C30 and QLQ-BN20) are analysed for long-term time points only (nominal Days 155 and 239). Analyses will be performed using a linear mixed effects model with subject-specific random intercept, including baseline value of the endpoint and a treatment-by-visit interaction. Model-based estimates by long-term treatment group and visit will be reported with 95% confidence intervals, together with pairwise differences versus the 0 mg long-term treatment group.

5.3.3. Steroid use

Steroid use is analysed as a dichotomous endpoint derived from the categorical response and will be analysed separately for the short-term and long-term periods. A baseline steroid indicator will be included as an adjustment variable.

5.3.3.1. Short-term analyses (randomised dose during short-term period).

For nominal Days 15, 29, and 43, a mixed logistic regression model with subject-specific random intercept will be fitted, including baseline steroid indicator, nominal visit (categorical), and randomised dose group. Results will be presented as model-based estimated risks by dose group with 95% confidence intervals, and as risk differences versus 0 mg with 95% confidence intervals.

5.3.3.2. Long-term analyses (randomised long-term treatment strategy).

For nominal Days 155 and 239, a logistic regression model including baseline steroid indicator and a treatment-by-visit interaction will be fitted. Inference will use variance estimation clustered by participant. Results will be presented as model-based estimated risks by long-term treatment group and visit with 95% confidence intervals, and as risk differences versus the 0 mg long-term treatment group with 95% confidence intervals.

5.3.4. RANO Response Endpoints

Ordinal tumour response endpoints defined according to the Response Assessment in Neuro-Oncology (RANO) criteria will be analysed at scheduled long-term follow-up assessments. Analyses will be performed separately by visit using proportional odds (cumulative logistic) models with treatment group as the explanatory variable. The population-level summary will be the proportional odds ratio comparing each dose level with no losartan, reported with 95% confidence intervals.

5.4. Secondary Biomarker Endpoints (Blood/Serum)

Blood/serum biomarker endpoints (Secondary Objective 3) are analysed analogously to the secondary imaging endpoints, separately for the short-term and long-term periods. All analyses are exploratory, performed on the Full Analysis Set, and no multiplicity adjustment is applied. Each analyte is analysed separately.

Short-term analyses (randomised dose during short-term period). For each analyte, a linear mixed effects model with subject-specific random intercept is fitted to the change from baseline on the natural logarithmic scale, including log baseline value of the analyte and randomised dose group as fixed effects, using the post-baseline samples within the short-term period (nominal

Days 15 and 43). Model-based estimated mean changes from baseline by dose group are reported with 95% confidence intervals, together with pairwise differences versus 0 mg.

Long-term analyses (randomised long-term treatment strategy). For each analyte, a linear model is fitted to the change from baseline at the long-term assessment (nominal Day 225) with long-term treatment group as the explanatory variable and baseline value as a covariate. Model-based estimated mean changes from baseline by long-term treatment group are reported with 95% confidence intervals, together with pairwise differences versus the 0 mg long-term treatment group. A mixed effects model is used in place of the linear model where more than one long-term sample per participant is available.

Scale of analysis: For each analyte, the change from baseline is analysed on the original (linear) concentration scale unless a log-transformation provides a better fit. The transformation is pre-specified to be selected per analyte by the Akaike Information Criterion (AIC) from the short-term model: the log scale is used when it yields the lower AIC, otherwise the linear scale is retained. To preserve comparability, the long-term analysis uses the same scale selected for the corresponding analyte in the short-term analysis. Where the log scale is used, a fixed per-analyte offset (half the smallest positive observed value) is added before transformation to accommodate non-positive values, and results are labelled as analysed on the log scale.

Missing biomarker data are handled implicitly through likelihood-based estimation under the missing-at-random assumption, consistent with Section 2.4. No explicit imputation is performed.

5.5. Secondary Time-to-Event Endpoints

Secondary time-to-event endpoints will be analysed using Cox proportional hazards models with long-term treatment dose as the primary explanatory variable. Hazard ratios comparing each active dose with no losartan will be reported with 95% confidence intervals. Proportional hazards assumptions will be assessed using standard diagnostic methods. Kaplan-Meier plots and overall log-rank test results will be provided as supportive displays. All analyses are exploratory and no MCP-Mod methodology will be applied.

5.6. Sensitivity and Supplementary Analyses

No formal sensitivity analyses beyond those specified in Section 1 are planned for secondary estimands. Supplementary descriptive summaries and graphical displays may be provided to aid interpretation.

6. Analyses Supporting Exploratory Objectives

Further exploratory objectives will not be considered in this analysis plan but specified in separate plans.

7. Safety Analyses

7.1. Analysis Sets

Safety analyses will be conducted using the Safety Set, defined as all participants who received at least one dose of losartan. Participants will be generally analysed according to the treatment dose received.

7.2. General Principles

All safety analyses are exploratory and descriptive in nature. No formal hypothesis testing is planned for safety endpoints.

Because of the stepped-wedge design, a participant's losartan dose changes over the course of the study, so adverse events are summarised against the dose to which exposure is attributed at the time of onset rather than against a single randomised dose. Two windows are distinguished:

Short-term window (onset on Day 44 or earlier): events are attributed to the stepped-wedge dose in effect on the day of onset. Events occurring before a participant's randomised step has begun are attributed to the 0 mg group. As a participant may contribute events at more than one dose, the dose groups in this window are not mutually exclusive; the denominator for each dose is the number of participants exposed to that dose at any time during Days 1–44.

Long-term window (onset on Day 45 or later): events are attributed to the participant's assigned long-term dose, with the denominator being the number of participants in each long-term dose group.

This attribution ensures adverse events are summarised against the exposure actually present when they occurred. Counts and proportions of participants are presented within each window; no formal between-dose comparison is made.

Adverse events and clinical safety parameters will be analysed using a treatment policy strategy, such that all observed safety data are included regardless of treatment discontinuation, treatment interruptions, or changes in background therapy.

7.3. Adverse Events

Adverse events will be coded using the Medical Dictionary for Regulatory Activities (MedDRA). The incidence of treatment-emergent adverse events will be summarised by treatment dose group, system organ class, and preferred term.

Summaries will include:

- Number and proportion of participants experiencing at least one adverse event
- Number and proportion of participants experiencing serious adverse events

- Number and proportion of participants experiencing adverse events leading to treatment discontinuation

Adverse events will be summarised overall and by maximum severity and relationship to study treatment, where available.

7.4. Vital Signs

Vital signs (systolic blood pressure, diastolic blood pressure, and pulse) will be summarised by treatment dose group and visit using descriptive statistics for observed values and changes from baseline.

In addition, to provide interpretable summaries of short-term and long-term safety effects, vital signs will be analysed using repeated-measures mixed effects models analogous to those used for continuous efficacy endpoints. Models will include baseline value of the parameter, nominal visit (categorical), and treatment group, with a subject-specific random intercept to account for within-participant correlation. Where relevant, terms accounting for time/period effects inherent to the stepped-wedge design will be included.

Short-term summaries will use post-baseline measurements within the short-term period; long-term summaries will use scheduled long-term follow-up assessments. Results will be presented as model-based mean changes from baseline by dose group with 95% confidence intervals. These analyses are supportive and will be interpreted descriptively.

7.5. Clinical Laboratory Parameters

Conventional laboratory blood parameters will be summarised by treatment dose group and visit using descriptive statistics for observed values and changes from baseline.

To support interpretation, laboratory parameters will also be analysed using repeated-measures mixed effects models as described for vital signs, including baseline adjustment, nominal visit, treatment group, and a subject-specific random intercept. Short-term and long-term summaries will be produced using the corresponding post-baseline assessments within each period. Results will be presented as model-based mean changes from baseline by dose group with 95% confidence intervals and interpreted descriptively.

Where relevant, values will additionally be categorised according to laboratory reference ranges, and the number and proportion of participants with values outside the normal range will be summarised descriptively.

7.6. Deaths

All deaths occurring during the study will be listed and summarised descriptively by treatment dose group, in addition to the overall survival analysis as described above

7.7. Extent of Exposure

The extent of exposure to losartan will be summarised descriptively to support interpretation of safety findings and to characterise adherence to the planned treatment schemes in this stepped-wedge trial.

Summaries will include:

- The number and proportion of participants exposed to each planned dose level of losartan (0 mg, 25 mg, 50 mg, and 100 mg)
- Duration of exposure to losartan overall and by dose level
- The number and proportion of participants who followed the planned short-term treatment scheme only and those who continued to long-term treatment
- The number and proportion of participants who switched dose levels or discontinued losartan prior to the planned end of treatment

Exposure duration will be summarised using descriptive statistics (e.g. mean, median, minimum, maximum) and categorical summaries (e.g. ≤ 14 days, >14 to ≤ 42 days, >42 days), overall and by dose group.

Where relevant, summaries will distinguish between:

- Randomised treatment assignment during the short-term period, and
- Actual treatment received during the long-term period

Graphical displays illustrating treatment exposure over time and transitions between dose levels may be provided to aid interpretation.

These summaries are descriptive and are not used to redefine the estimands or analysis populations, but are intended to provide context for the safety and efficacy analyses.

8. Other Analyses

8.1. Other Variables and/or Parameters

Not applicable

8.2. Subgroup Analyses

The number of observations does not allow for sensible subgroup analyses in this trial.

9. Interim Analyses

Not applicable, no interim analyses planned or performed in this trial.

10. Changes to Protocol-planned Analyses

10.1. Multiplicity and MCP-Mod Application

In the SAP, the MCP-Mod methodology is **restricted to the primary imaging endpoints**, for which it was originally intended. Time-to-event endpoints are analysed using Cox proportional hazards models without MCP-Mod. This change reflects methodological appropriateness for the endpoint type and does not affect the study objectives.

In addition, the SAP specifies that MCP-Mod hypothesis tests are conducted as **one-sided tests at a nominal 2.5% significance level**, consistent with standard MCP-Mod methodology. This represents a clarification relative to the protocol text.

10.2. Specification of Candidate Dose–Response Models

The protocol refers to a “step model” as one of the candidate dose–response models. In the SAP, this is operationalised using a **piecewise linear (linear interpolation) model with knots at the planned dose levels**, which allows step-like dose–response patterns without imposing smoothness assumptions beyond continuity. This represents a clarification of the mathematical implementation of the protocol-specified model.

10.3. Supplementary Analysis of Any Dose Versus No Dose

The SAP includes a supplementary analysis comparing **any dose of losartan** (25 mg, 50 mg, or 100 mg pooled) versus **no losartan (0 mg)** using the primary mixed effects model. This analysis was not explicitly specified in the protocol and is added to provide a descriptive assessment of a potential common treatment effect corresponding to a step-like dose–response pattern. This analysis is supplementary and not part of the multiplicity-controlled testing strategy.

10.4. Summary

All changes described above were made prior to database lock without knowing any comparative results. None of the changes alter the study objectives or primary estimands. The modifications primarily clarify implementation details, improve alignment with the estimand framework, and ensure appropriate methodology for different endpoint types.

11. Sample Size Determination

Copied from the protocol:

The underlying repeated measures mixed model is defined as:

$$\Delta P_{ij} = A_i + \mu_{trt} + \mu_j + \epsilon_{ij}$$

where ΔP_{ij} is the change from the MRI-based baseline Perfusion value, measured as a percentage of the optimal value (100%) of patient $i = 1..N$ at timepoint j , A_i is the random intercept of patient i , μ_{trt} is the treatment effect, μ_j is the expected P_{ij} at timepoint j , and ϵ_{ij} is the residual. Note that we assume the treatment effect is equal between the treatment doses, such that the μ_j denotes the average treatment effect over the three doses. We assume the effect size to be 20%, however, we also assess the power of alternative effect sizes of 5%, 10% and 15%. The variance is difficult to assess, but we assume an overall standard deviation of 15%⁸⁹, split with a within- and between patient standard deviation of 10.6%. We used simulations to estimate the sample sizes and corresponding powers separately for each of the three indications.

11.1. Simulations

A total of 1000 simulated trials were simulated for each sample size. For each simulation we estimated the parameters of interest (treatment effect) and the p-value from a mixed model. The power is calculated as the proportion of trials with significant treatment effect. For each simulated trial we simulate 6, 12 or 18 patients in each intervention group (step). In the calculations, the doses are pooled to assess the overall treatment effect of losartan.

Sample size per group	Effect size	Total sample size	Power	Estimate
6	5	18	0.187	4.822
6	10	18	0.649	10.214
6	15	18	0.933	15.028
6	20	18	0.998	20.225
12	5	36	0.353	4.917
12	10	36	0.921	10.215
12	15	36	0.999	14.973
12	20	36	1.000	19.987
15	5	45	0.437	5.047
15	10	45	0.961	10.070
15	15	45	1.000	15.007
15	20	45	1.000	19.843
18	5	54	0.498	4.931
18	10	54	0.979	10.057
18	15	54	1.000	14.968
18	20	54	1.000	19.995

* Power calculations shown in bold font are assumed of sufficient magnitude.

Following the simulations, we see that with an effect size of 20% we have very high power for all sample sizes. With 15 patients in each of the indications, we see that we gain a sufficient power even when the effect size is as low as 10%.

As a result, we need 45 completed patients in each of the two glioblastoma indications, and 36 with brain metastases. To adjust for a possible 15-20% drop-out (non-compliance and mortality), we need to randomize 54 patients in each of the glioblastoma indications and 45 with brain metastases, to a total of 153 patients.

In addition to looking at the primary biomarker endpoint, we want to assess the long-term effect of the intervention on clinical endpoints. The purpose of this long-term follow-up is exploratory (hypothesis generating), as a properly powered hypothesis testing trial on e.g. overall survival would require a lot more patients. However, to have any power to generate hypotheses on clinical endpoints we will randomize half of the patients on 25 mg and 50 mg to either continued or discontinued losartan treatment (same dose), while patients on 100 mg will all continue losartan treatment until day 168 (Study AR) or 238 (Study AN). Following this design, we will end up with a distribution of 18:9:9:18 with no treatment, 25 mg, 50 mg and 100 mg, respectively. The 2:1:1:2 distribution has more power to show a dose-response relationship compared to a standard 1:1:1:1 distribution.

12. Supporting Documentation

The following documents and resources support the implementation, transparency, and reproducibility of the statistical analyses described in this SAP.

12.1. Mock-up Statistical Report

A mock-up statistical report has been prepared to illustrate the planned tables, figures, and listings for the primary and secondary analyses. The mock-up report is generated using simulated or sham-randomised data to avoid disclosure of treatment effects prior to database lock and unblinding.

The mock-up report is intended solely to demonstrate the structure, content, and presentation of the final statistical outputs and does not represent results from the actual trial data.

12.2. Statistical Programming Code

The statistical analyses described in this SAP are implemented using reproducible statistical code maintained in a version-controlled repository:

<https://github.com/ocbe-uio/impress>

The repository contains scripts for data preparation, analysis, and generation of tables and figures, as well as configuration files supporting reproducible execution of the analyses.

All main components of the statistical analysis code relevant to the primary and secondary analyses will be **finalised, reviewed, and version-locked prior to database lock**. Any changes to the analysis code after database lock will be documented, justified, and traceable.

12.3. Use of Sham Randomisation

During development and validation of the statistical analysis code and mock-up report, analyses are performed using sham randomisation or simulated treatment assignments. This ensures that the analysis pipeline, model specifications, and output structure are fully validated without revealing results based on the true randomised treatment allocation.

Final analyses will be conducted only after database lock, using the true randomisation, and strictly according to the methods specified in this SAP.

12.4. Reproducibility and Open Science Framework Repository

To support transparency, reproducibility, and long-term accessibility of study documentation, an independent project repository will be established on the **Open Science Framework (OSF)**.

The OSF repository will serve as a curated entry point and will include, or reference, the following materials:

- The final study protocol and protocol amendments
- This Statistical Analysis Plan and subsequent SAP versions
- The mock-up statistical report
- Links to the version-controlled statistical code repository
- Additional supporting documentation relevant to the statistical analyses

The OSF repository will not supersede the SAP as the governing document for the statistical analyses, but will provide a transparent and traceable framework for accessing supporting materials.

12.5. Compliance with Clinical Trials Unit Procedures

All statistical analyses will be performed in accordance with the procedures and quality standards of the Oslo University Hospital Clinical Trials Unit (CTU), as described in the institutional quality handbook:

<https://ehandboken.ous-hf.no/document/131441>

These procedures cover data handling, version control, quality assurance, documentation, and traceability of statistical analyses.

13. Appendices

13.1. Primary model fit

The model fit for different covariance structures in the primary statistical model used in ImPRESS is given in the “ImPRESS Model selection” document.

13.2. Proportional hazard model fit

The pre-unblinding assessment of the proportional-hazards (PH) assumption for the Cox models reported in the time-to-event section is given in the “mPRESS — Time-to-Event Proportional-Hazards Assessment” document.

13.3. Biomarker model fit

The pre-unblinding assessment of model fit for the 33 biomarker analytes from Bio-Plex19-plex and 14-plex serum panels are given in “mPRESS — Biomarker Model Fit Assessment”.

Participants

OSLO UNIVERSITY HOSPITAL Norway

Signed with E-sign

2026-06-24 14:08:04 UTC

Signatory

Date

Inge Christoffer Olsen
i.c.olsen@medisin.uio.no

Delivery channel: Email
IP Address: 84.208.208.163

PETTER BRANDAL Norway

Signed with E-sign

2026-06-24 14:08:47 UTC

Signatory

Date

Petter Brandal
pebra@ous-hf.no

Delivery channel: Email
IP Address: 91.186.70.20

KYRRE EEG EMBLEM Norway

Signed with E-sign

2026-06-24 15:12:32 UTC

Signatory

Date

Kyrre Eeg Emblem
kemblem@ous-hf.no
+4797080018

Delivery channel: Email
IP Address: 91.186.70.21